

DPhil Thesis Draft



Alessandro Lodi
Trinity College
University of Oxford

A thesis submitted for the degree of
Doctor of Philosophy in Materials

Michaelmas 2021

Contents

List of Figures	vii
Abbreviations	ix
1 Introduction	1
1.1 Motivation	1
1.2 Graphene	1
1.2.1 crystal structure	1
1.2.2 band dispersion	3
1.2.3 Graphene Transistors	5
1.3 Carbon-based Field Effect Transistor	12
1.3.1 crystal structure	12
1.3.2 Carbon Nanotube Field-Effect Transistors	13
1.4 Graphene Nanoribbons	20
2 Foundations	27
2.1 Charge Transport In Zero-Dimensional Objects	27
2.2 Charge Transport in One-Dimensional Objects	27
2.3 Thermoelectric Phenomena in Zero-Dimensional Structures	28
2.4 Density Functional Theory	28
2.5 Non-Equilibrium Green Function Theory	28
3 Instrumentations	29
3.1 Room Temperature DC Measurements	29
3.2 Dipstick DC Measurements	29
3.3 Cryogenic DC Measurements	29
3.4 Cryogenic AC Measurements	30
3.5 Thermoelectric Measurements	30
4 Fabrication	31
4.1 General Fabrication Procedure	31
4.1.1 Graphene Preparation and Wet Transferring	31
4.2 Graphene Nanogaps	32

4.3	Chip Design	32
4.3.1	2T Design	32
4.3.2	3T Design	32
4.3.3	Thermoelectric Device	32
4.4	Deposition Process and Related Phenomna	32
5	Room Temperature Graphene Nanoribbon Single-Electron Transistor	33
5.1	Introduction	33
5.2	Sample Preparation and Characterisation	33
5.3	Electrical Transport Measurements	34
5.4	Summary	34
6	Chemical Doping of the Bandgap for Cove Graphene Nanoribbons	35
6.1	Introduction	35
6.2	Sample Preparation and Characterisation	35
6.3	<i>Ab-initio</i> Experiments	35
6.4	Electrical Transport Measurements	36
6.5	Summary	36
7	Thermocurrent Spectroscopy of cove Graphene Nanoribbons	37
7.1	Introduction	37
7.2	Sample Preparation and Characterisation	37
7.3	Electrical Transport Measurements	38
7.4	Summary	38
8	Transport Spectroscopy of Biaryl-linked Dy–Tb Porphyrin	39
8.1	Introduction	39
8.2	Sample Preparation and Characterisation	39
8.3	Electrical Transport Measurements	40
8.4	Summary	40
9	Conclusions and Outlook	41
9.1	Conclusions	41
9.2	Outlook	41
Appendices		
A	Review of Cardiac Physiology and Electrophysiology	45
A.1	Anatomy	45
A.2	Mechanical Cycle	48
A.3	Electrical Cycle	49
A.4	Cellular Electromechanical Coupling	51

List of Figures

1.1	Graphene density of states plotted over a wide range of nearest neighbor hopping parameter t , showing pronounced van Hove singularities for $\epsilon \sim \pm t$ [1]. The dashed rectangle indicates the almost linear low energy DOS normally spanned in back-gated transport measurements.	5
1.2	Majorana T-junction. Band gap versus nanoribbon width from experiments [59, 62–64] and theoretical calculations [65, 66]. The following abbreviation are used: ac for <i>armchair</i> , zz for <i>zigzag</i> . Figure adapted and extended from [25]	21
1.3	(a) E_g vs W for four (P1–P4) of the parallel devices and two (D1, D2) with varying orientation. The inset shows E_g vs relative angle θ for the device D1 and D2. (b) E_g^{-1} vs W . (c) Conductance vs width of parallel GNRs measured at a gate voltage $V - V_g = 50$ V. Squares are values measure at 300 K whereas triangle corresponds to 1.6 K. The insets show the conductivity (upper) and the inactive GNR width (lower) obtained from the slope and x intercept of the linear fit at varying temperatures. Adapted from [62].	26

Abbreviations

- CMOS** complementary metal-oxide-semiconductor. 22
- CNT** carbon nanotube. 12, 13, 14, 20
- CNTFET** carbon nanotube field-effect transistor. 13, 14, 16, 18
- DFT** Density Functional Theory. 19
- DIBL** drain induced barrier lowering. 16
- EOT** effective oxide thickness. 17, 18
- FET** field-effect transistor. 6, 13
- GNR** graphene nanoribbon. 20
- GNRFET** graphene nanoribbon field-effect transistor. 24
- MOSFET** metal-oxide-semiconductor field-effect transistor. 9, 10, 11, 17
- MW** multi-walled. 12
- NEGF** non-equilibrium Green function. 19
- SB** Schottky barrier. 14
- SS** Sub-threshold swing. 16, 17, 18
- SW** single-walled. 12

1

Introduction

1.1 Motivation

Motivation

1.2 Graphene

1.2.1 crystal structure

Graphene has a hexagonal structure where the carbon atoms are hybridised sp^2 , as detailed by the Pauling theory on chemical bonds. Based on the experimental evidence collected in the last 20 years, the model that seems to best describe the proprieties of graphene is the band-structure Wallace model, which dates back in 1947, and describes graphene as a semimetal with a linear dispersion of the electronic excitations. In the International Table of Crystallography, graphene is included in the symmorphic space group #191, with symmetry D_6^1h ($P6/mmm$). The triangular lattice is obtained by considering two atoms per unit cell having the respective lattice vectors:

The carbon atoms in graphene form a two-dimensional hexagonal crystal with two atoms per unit cell, as shown in figure.

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}), \quad \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \quad (1.1)$$

with $a \approx 1.42 \text{ \AA}$ be the carbon-carbon bond length. Each atom belonging to one sublattice, called sublattice A, has three first neighbours belonging to the other sublattice, called sublattice B. With respect to one atom chosen from sublattice A as origin, the position of these first three neighbours in real space are given by:

$$\boldsymbol{\delta}_1 = \frac{a}{2}(1, \sqrt{3}), \quad \boldsymbol{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}), \quad \boldsymbol{\delta}_3 = -a(1, 0), \quad (1.2)$$

The reciprocal-lattice vectors are then given

$$\mathbf{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}), \quad \mathbf{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}), \quad (1.3)$$

In order to define the vertices of the Brillouin zone one has to apply the Bragg condition:

$$2\mathbf{k} \cdot \mathbf{G} = 0 \quad (1.4)$$

where

$$\mathbf{G} = n\mathbf{b}_1 + m\mathbf{b}_2, \quad (1.5)$$

The first amongst these are:

$$\begin{array}{lll} n = \pm 1, & m = 0, & \mathbf{G} = \pm \mathbf{b}_1, \\ n = 0, & m = \pm 1, & \mathbf{G} = \pm \mathbf{b}_2, \\ n = \pm 1, & m = \pm 1, & \mathbf{G} = \pm(\mathbf{b}_1 + \mathbf{b}_2), \end{array}$$

and all these vectors have the same modulus $|\mathbf{G}| = \frac{4\pi}{3a}$. By applying Bragg's condition in the first of the three cases we get:

$$\sqrt{3}k_x + k_y = \mp \frac{4\pi}{3a}, \quad (1.6)$$

which gets satisfied by the points:

$$K \equiv \mp \frac{2\pi}{3a} \left(\frac{1}{\sqrt{3}}, 1 \right) \quad (1.7)$$

For the second case one gets:

$$\mp (\sqrt{3}k_x - k_y) = \frac{4\pi}{3a}, \quad (1.8)$$

which gives the points:

$$K' \equiv \pm \frac{2\pi}{3a} \left(\frac{1}{\sqrt{3}}, -1 \right) \quad (1.9)$$

The third and last case gives the point:

$$M \equiv \pm \left(0, \pm \frac{2\pi}{3a} \right) \quad (1.10)$$

1.2.2 band dispersion

The electronic structure of graphene can be model with a tight-binding hamiltonian. Following the second quantization formalism ($\hbar = 1$ units), the hamiltonian can be written as:

$$\mathcal{H}_{\text{TB}} = -t \sum_{\langle ij \rangle, \sigma} \left(a_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.} \right) - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} \left(a_{\sigma,i}^\dagger a_{\sigma,j} + b_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.} \right), \quad (1.11)$$

The notation $\langle ij \rangle$ means that the sum runs over pairs of electrons which are nearest neighbour; similarly, the notation $\langle\langle ij \rangle\rangle$ specify a summation over the next nearest neighbours; the operators $a_{i,\sigma}^\dagger$ ($a_{i,\sigma}$) and $b_{i,\sigma}^\dagger$ ($b_{i,\sigma}$) creates (annihilates) an electron with spin σ ($\sigma = \uparrow, \downarrow$) for atoms on sublattice A and B, respectively. The value of t which reproduces best experimental result is $t \approx 2.8$ eV and quantify the nearest-neighbour hopping energy; similarly, t' denotes the next nearest-neighbour hopping energy. The dispersion energy computed from this model reads:

$$E_{\pm}(\mathbf{k}) = \epsilon_p - t' f(\mathbf{k}) \pm t \sqrt{3 + f(\mathbf{k})}, \quad (1.12)$$

with $f(\mathbf{k})$ defined as:

$$f(\mathbf{k}) = 2 \cos(\sqrt{3}k_y a) + 4 \cos\left(\frac{\sqrt{3}}{2}k_y a\right) \cos\left(\frac{3}{2}k_x a\right). \quad (1.13)$$

Therefore, it is possible to identify an upper (π^*) band associated with the plus sign and a lower (π) band described by the minus sign; these are symmetric with respect to the single-particle zero energy. The experimental estimate reported for next nearest-neighbour hopping energy is $t' \approx 0.1\text{eV}$ thus the asymmetry introduced is small. It is worthwhile to notice that the function $3 + f(\mathbf{k})$

If the system is not doped, there is one electron per carbon atom, so two electrons per cell: it follows that the lower band is full and the upper band is empty the upper one is empty. The system is therefore a semi-metal. The peculiarities of the dispersion in graphene become apparent when analysing their trend near the point $\mathbf{K}$ and $\mathbf{K}'$. By taking the positive components for the $\mathbf{K}$ as define in Eq. 1.7 and expanding the Eq. 1.12 as $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|\mathbf{q}| \ll |\mathbf{K}|$, one gets:

$$E_{\pm}(\mathbf{q}) = \epsilon_p \pm \frac{3at}{2}|\mathbf{q}| \quad (1.14)$$

which is linear in $\mathbf{q}$. For intrinsic graphene, at $T = 0\text{K}$ $E_F = \epsilon_p$. The Fermi velocity is defined by the usuale relation $v_F = \frac{1}{\hbar} \left(\frac{dE}{d\mathbf{k}} \right)$, and in graphene $v_F \simeq 1 \times 10^6 \text{m s}^{-1}$.

A difference arises in the density of the states, when we limit ourselves to consider those near the Fermi energy. In the case of a two-dimensional electron gas, it is constant. On the contrary for graphene, the density of states per unite cell is:

$$\rho(E) = \frac{2A_{\text{cell}}}{\pi} \frac{|E|}{v_F^2} \quad (1.15)$$

If the graphene is charge-neutral, the DOS has two Van Hove singularities at $\omega_{\text{VH}} = \pm 1\text{eV}$ [2]. This behaviour is in stark constrat with the DOS of carbon nanotubes and clean graphene nanoribbons, as they show $1/\sqrt{E}$ singularities due to the 1D nature, which imposes a quantization perpendicular to the ribbon or tube longitudinal direction.

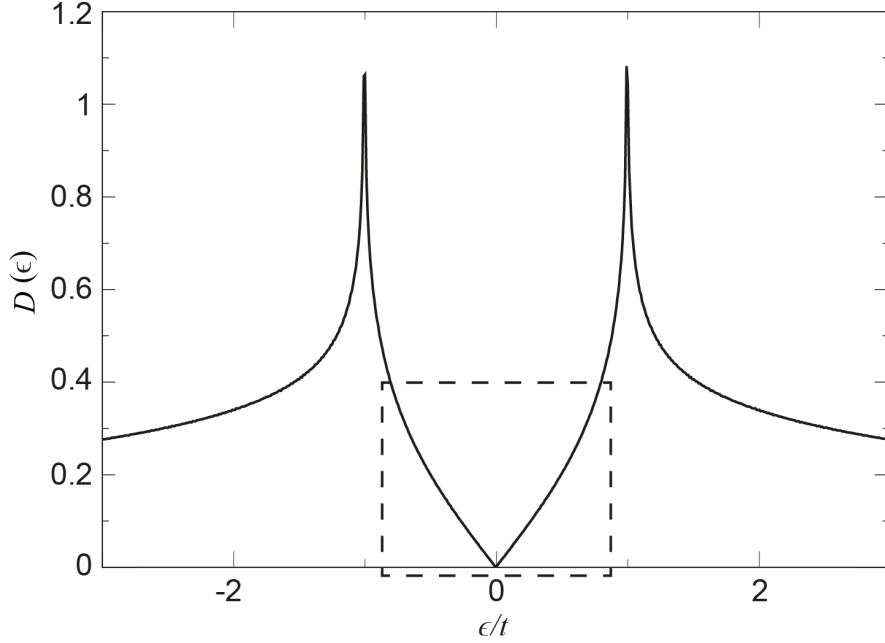


Figure 1.1: Graphene density of states plotted over a wide range of nearest neighbor hopping parameter t , showing pronounced van Hove singularities for $\epsilon \sim \pm t$ [1]. The dashed rectangle indicates the almost linear low energy DOS normally spanned in back-gated transport measurements.

1.2.3 Graphene Transistors

For transistor applications, a material needs a highly conductive on-state and an off-state where the source-drain current I_{sd} does not flow. For RFs applications, on the other hand, the full speed operation is obtained by operating in the (drain) current saturation regime. Both applications requires a sizeable band gap. Furthermore, the IC scaling trend is often juxtaposed to chip-level issues such as power densities (currently on the order of 1000 W/cm^2), heat generation, and chip temperatures reaching levels that will prevent reliable operations.

In CMOS logic, the charge carriers are accelerated by the electric field $\mathcal{E}_{sd}$ between source and drain, thus undergoing five possible sources of scattering: electron–electron scattering, electron–phonon (lattice vibrations), material interfaces, imperfections or impurity atoms. The first two are more fundamentals, while the last three are material-related. The most dominant mechanism through which charge carriers lose net energy is by scattering with phonons, consequently heating

up the lattice via Joule heating. For RF applications, on the other hand, heat is produced by the operational frequency at which the carriers operate. Thus, two further requirements are needed: on one hand a good thermal conductivity, but also small heat transfer with the substrate. Clearly trade-offs are here necessary.

The 2020 International Roadmap for Devices and Systems - which has replaced since 2017 the International Roadmap for Semiconductors - dedicates ample attention to graphene in its report *Beyond CMOS*, in which technologies and semiconductor materials to complement and replace Si are considered. The race to scale the channel in a field-effect transistor (FET) still motivates a significant effort on introducing high-mobility channel materials, since the carrier mobility is an appropriate parameter to address carrier transport even in nanometer FETs [3]. In fact, most of the early excitements in using graphene as channel material in a standard FET technology were sparked by considering the outstanding mobility values reported since its discovery.

At 290 K, the intrinsic mobility μ_e^{in} of graphene is exceptionally high. The first data were the one obtained in 2004, where graphene on SiO_2 displayed an electron mobility $\mu_e^{\text{in}} \approx 10,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ [4]. It spans $\mu^{\text{in}} \sim 3000 - 12000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ in the concentration range $n \sim 1 - 0.1 \times 10^{12} \text{ cm}^{-2}$. Employing Einstein's relation, the scattering time of about:

$$\tau = \frac{\hbar\sigma(\pi/n)^{1/2}}{e^2v_F} \sim 2\text{ns} \quad (1.16)$$

a mean free path on the range:

$$l = v_F\tau \sim 55\text{-}75\text{nm} \quad (1.17)$$

at room temperature.

The mobility is limited by different factors such as scattering by charged impurities or defects and electron-phonon scattering. In very clean and suspended samples (where the underling SiO_2 substrate has been etched away) the mean free path l increases by almost two order of magnitude, becoming comparable to the device size, an indication that the ballistic transport regime is reached

The slope of the *v-shaped* transfer curve gives the low-field *field effect mobility*¹ [Meric2008] $\mu^{\text{FE}} = (1/C_{\text{gate}})(\partial\sigma/\partial V_{\text{gs}})$ that is constant over a quite wide range of n (yielding in this case $\mu_{\text{h}}^{\text{FE}} \sim 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_{\text{e}}^{\text{FE}} \sim 1800 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for the low n region).

. Later in 2008 [5] and 2013 [6], higher mobility values were reported: $\approx 120,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at 240 K and $\approx 1,000,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at 4 K. The reason for these higher values was attributed to the use of suspended graphene rather onto substrate, where surface effects inevitably affect the carrier dynamics. At the same time, results ($\approx 120,000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) obtained at room temperature with graphene onto a single-layer of hexagonal boron nitride (hBN) [7] and sandwiched between two hBN layer [6] have suggested that this flat and inert materials can be act as a excellent passivation film for graphene devices.

One of the figure of merit often taken into account for addressing logic applications for a materials is the ratio between the on-state current and the off-state current. The so called on-off ratio $I_{\text{on}}/I_{\text{off}}$ has to be in the range 10^4 – 10^7 . While the I_{on} is not directly related to the band gap E_g , it is possible to link the value of I_{off} with the E_g .

In an metal–semiconductor junction (M/S), the current is built by the majority of the carriers, these being electrons or holes for n -type or p -type semiconductors. Most of the graphene–semiconductor junctions formed during experiments have typical dopant concentration $N \leq 10^{17} \text{ cm}^{-3}$. The $I - V$ characteristics of M/S junctions with such dopant concentrations are well reproduced by the thermoionic emission theory. If we then exclude other possible conduction mechanisms (standard and thermoionic field emission, generation/recombination, diffusion of electron, injection of holes and leakage current), the I_{off} becomes proportional to the barrier the electrons have to overcome from the semiconductor to the metal, namely

$$I_{\text{off}} \propto \exp\{(e\beta\phi_B)\}. \quad (1.18)$$

¹to not be confused with the *intrinsic* Drude mobility $\mu^{\text{in}} = \sigma/ne$ or the Hall mobility $\mu_e^H = -\sigma_e R_H$.

	Direct Gap			Indirect Gap		
	InP	Si	Ge	GaAs	InAs	Graphene
a_{lattice} (Å)	5.87	5.43	5.65	5.65	6.06	2.46
E_g (eV)	1.35	1.12	0.66	1.42	0.36	0
μ (cm ² V ⁻¹ s ⁻¹) at 300 K	2800	1400	3900	4600	16,000	200,000
v_{sat} (10 ⁷ cm s ⁻¹)	2.2	1	0.6	2.2	4.0	>5
DOS (m^*/m_0)	0.077	1.08	0.56	0.067	0.023	0
ϵ_r	12.4	11.9	16.0	13.1	14.6	2.4
κ (W m ⁻¹ K ⁻¹)	68	150	60.2	46	27	5000

Table 1.1: Differences between the main semiconductors used in modern digital electronics and graphene.

where e is the electron charge, ϕ_B is the Schottky barrier height, and $\beta = k_B T$. If we assume that the work function of graphene and the metal are the same, then $\phi_B = E_g/2$, where E_g is the band gap energy originating from the difference of the work function of graphene and the metal; thus, the on–off ratio can be estimated from the useful proportionality [8]:

$$\frac{I_{\text{on}}}{I_{\text{off}}} \propto \exp\left\{\frac{E_g}{2k_B T}\right\}, \quad (1.19)$$

where the thermal energy $k_B T$ and the bandgap are both expressed in meV. While for high frequency applications the on–off ratio is of minor importance, several references in literature report suitable values for CMOS bandgap should fall in a range between 300 and 500 meV [9, 10].

Hysteresis between the forward and reverse sweeps have been reported depending on the sweeping rate and range [11], and attributed mainly to charge trapping in silanol groups (Si–OH), mediated and assisted by surface-bound H₂O molecules [12]. Despite the usefulness for its proof-of-concept purposes, back-gate devices

suffered from parasitic capacitances, making impossible further integration with other components. Alternative geometries have since then been explored, in order to increase the carriers mobility, to decrease contact resistances and parasitic capacitances, and to avoid Schottky barriers formation between the graphene and the semiconductor. Top-gate devices have been the first to chronologically appear in the literature. Back then in 2007, Lemme and co-workers used a stack of SiO₂ (20 nm), Ti (10 nm), and Au (100 nm) as top gate for modulating the current inside a large-area graphene channel transistor obtained by the exfoliation method [13]. Whilst the gate characteristics exhibit a similar trend of the characteristics (the top gate reduces the back-gate current modulation, lowering the drain current by one order of magnitude) and the hysteresis, weak current saturation at high drain-source bias have been reported [14]. This effect seems due to a partial charge pinning at the contacts at high back-gate voltages [15] and carrier drift velocity eventually saturating due to optical-phonon scattering [16]. Top-gated graphene metal–oxide–semiconductor field-effect transistors (MOSFETs) with CVD and epitaxial graphene have been also made [17, 18]; SiO₂, Al₂O₃, and HfO₂ have been used as top-gate dielectric.

However, despite the improvements obtained with a top gate, most graphene MOSFET with large-area graphene channels show on–off switching ratios of less than 10 at room temperature [9], much too little for complex logic circuits.

In 2012, Mehr *et al.* proposed a vertical arrangement of emitter, base, and collector similar to the original vacuum triode, where graphene plays the base role and electrons travel vertically rather than transversely. When the device is in the ON state, the electrons tunnel with a Fowler-Nordheim tunnelling mechanism:

$$I \propto \left(\frac{V_{sd}}{d} \right)^2 \exp \left\{ -\frac{\alpha d}{V_{sd}} \right\}, \quad (1.20)$$

through the emitter–base layer (of thickness d); ON/OFF ratios of the order of 10⁴ have been demonstrated using graphene between two h-BN layers [19] and using WeS₂–graphene [20] heterostructures.

Solution methods have also been explored, in which graphene is dropcasted on top of a semiconducting wafer, and then spin-coated [21]. The main difficulties arise when attempting to quantify the structure and the quality of the deposited graphene, and not many methods can guarantee a good dispersion in organic solvents [22]. Simpler ways to obtain graphene oxide in solution have been reported [23], but once deposited, graphene oxide needs to be reduced for obtaining pure graphene – low mobility values (up to, at best, $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) caused by defects during the fabrication steps have discouraged further investigations along this direction.

Yet in spite of the very different dc behavior compared to the most used semiconductors in digital electronics, graphene FETs have reported good performance in terms of high-frequency applications. This comparison is made by measuring each system's frequency response, its output signal strength when it is given a particular input signal: features such as ballistic transport, linear current-voltage characteristics, high transconductance, and very high ($> 10^8 \text{ A cm}^{-2}$) sustainable currents are all ideal features for analog/RF applications [16, 24].

The primary means of estimating frequency response is through the evaluation of two transistor characteristic frequencies, namely cutoff f_T and maximum of oscillations f_{max} . The former indicates how quickly a signal can travel from the gate to the drain of the device, and it is usually determined when the small-signal current gain drops to unity, while the latter is measured when the power gain decreases to unity [9]. The frequency response depends inversely on gate length and proportionally to the transconductance, which is loosely proportional to mobility. Thus, reducing the gate length and using higher mobility materials are valid approaches to increase the device operational speed [25]. In this context, many research groups have successfully demonstrated RF capabilities with gapless graphene channels MOSFETs.

The first device with cutoff frequency f_T in the GHz range was reported in late 2008, and was fabricated by exfoliated graphene deposited on top of high-resistivity Si substrate, with 30-nm-thick top-gate HfO_2 dielectric and 500 nm of gate length

[24]. Despite a measured $f_T \approx 14.7$ GHz, it was argued that large contact resistances and gate length were the main responsible of poor performance $f_{\max} < 1$ GHz.

The following year, Pd/Au contacts and 12-nm-thick Al_2O_3 were employed in a 350-nm-gate [26]. While the cutoff frequency improved up to $f_T \approx 50$ GHz, little gains in terms of $f_{\max}$ were obtained of only 2 GHz. In general, limited performance in $f_{\max}$ represents a major issue as for most RF applications f_T is less important than $f_{\max}$.

It was only after two years that an increase of one order of magnitude both in the cutoff and the maximum frequency of oscillation was demonstrated [18]. The device was realised with epitaxially-grown on silicon carbide (SiC) graphene MOSFET having 240 nm of gate length, with Ti/Pd/Au for contacts and 10-nm-thick HfO_2 of insulating layer on top gates. Highest frequency as well as $f_T \approx 100$ GHz and $f_{\max} \approx 14$ GHz were measured, exceeding any device then present. The authors also argued that since $f_{\max}$ is strongly correlated to device layout and parasitic resistances originating from the metallic contacts, viable ways to improve this parameter would be a partial passivation of the metal contacts and a multi-finger gate layout

These performance were soon followed later that year by a 144-nm-length gate device with $f_T \approx 300$ GHz using mechanically exfoliated graphene and $\text{Co}_2\text{Si-Al}_2\text{O}_3$ nanowire as self-aligned multi top-gates [27]. This result was also corroborated by comparing it with the projected value using the relation that links dc transconductance and gate capacitance with the cutoff frequency when total input parasitic capacitance is negligible, namely $f_T = g_m/(2\pi C_g)$ [28]. The authors did not report the power $G_{\max}^{1/2}$ but only the current $|h_{21}|$ gain, thus hampering an evaluation of $f_{\max}$, but in 2012 a similar proposal was advanced by Cheng *et al.*, where Au/Ti/ Al_2O_3 gate stacks were employed in self-aligned transistors [29]. Despite the improvements reported ($f_T \approx 427$ GHz for a channel length of 67 nm) respect to CVD graphene transistors with comparable channel length, $f_{\max}$ decreased down to 8 GHz when the length was shrunk down to 46 nm, showing the key role played by gate and parasitic resistances at high frequency.

To further improve the power gain, works on the saturation behavior of short channel graphene transistor were also conducted by IBM in 2012, by using CVD graphene grown on a Cu foil and then transferred on diamond-like and SiC substrates [30]. These are appealing because high-quality wafers are available, thus making a scale up of graphene arrays relatively straightforward. Since the work function of graphene is $\phi^g = 4.5$ eV and $\phi^{4\text{H-SiC}} = 3.7$ eV, this means that $\Delta\phi^{g-4\text{H-SiC}} = 0.8$ eV, thus n -doping graphene [31]. Very cleverly, the authors then used plasma-enhanced CVD silicon nitride as an intrinsic p-type gate dielectric, counterbalancing the n -doped graphene channels. Encouraging performance at high speed were not only measured in good cutoff frequencies ($f_T \approx 350$ GHz), but also in a factor of 2 improvement of $f_{\max}$ (≈ 40 GHz). To the best of my knowledge, the current graphene transistor power gain record of about 70 GHz was obtained in 2013 by using (000-1) carbon-terminated surface of SiC instead of the (0001) Si one, indicating the contact resistance R_c between Pd/Au contacts and graphene as key responsible for the high $f_{\max}$.

While these results have aligned graphene MOSFET with the main solid-state competitors – such as GaAS pHEMT, InP HEMT and Si MOSFET – in terms of f_T , the performance in terms of $f_{\max}$ are off by about one [32, 33] or two orders of magnitude [34]. In literature, one of the often cited fundamental reason for this behaviour is the poor drain saturation current in graphene, which ultimately depends on the missing bandgap of the channel. Graphene-derived materials such as nanoribbons and carbon nanotubes MOSFET could then, given their intrinsic semiconducting nature, represent a possible solution to this problem.

1.3 Carbon-based Field Effect Transistor

1.3.1 crystal structure

This crystalline form of carbon can be considered as a rolled up sheet of graphene and comes in two major flavours: single-walled (SW) and multi-walled (MW). Carbon nanotubes (CNTs) display an electronic band gap which depends on the chirality and the diameter of the nanotube [35]. Having defined two unit vectors in

the real space, $\mathbf{a}_1$ and $\mathbf{a}_2$, a CNT can be described by the linear combination of these with integers coefficients (m, n) . Based on the values of m and n , CNTs can be metallic, quasi-metallic ($m - n$ is a multiple of 3, with $m \cdot n \neq 0$ and $m \neq n$), and semiconducting for the majority of the remaining cases – exceptions due to the curvature (spin–orbit couplings) in small-diameter tubes are also known [36]. Thus, a quantization of the electronic states in the circumferential direction is resulted when the nanotube diameter d_{CNT} is between 0.8 and 2.5 nm. Each of these substates is called subband and are determined by the periodic boundary condition. Having defined a circumferential wave vector $\mathbf{k}_c$ such that $\mathbf{k}_c \cdot \mathbf{c} = k_c |c| = k_c \pi d_{\text{CNT}} = 2\pi\nu$, for ν integer, it is possible from this relation to draw parallel lines in the first Brillouin zone of graphene. If the subband intersect one of the $\mathbf{K}$ points, then the resulting nanotube will have no energy gap and thus exhibit metallic properties (this condition is realised by having $m = n$ for the coefficients of the chiral vector). If none of the subbands pass through $\mathbf{K}$ then the resulting CNT will be semiconducting, having a E_g proportional to the smallest distance between a $\mathbf{K}$ point and the subband lines:

$$E_g = 2 \left(\frac{\partial E}{\partial k} \right) \frac{k_c}{3} = \frac{2}{3} \hbar v_F \frac{2\pi}{|c|} = \frac{4}{3} \hbar v_F \frac{1}{d_{\text{CNT}}} \approx \frac{0.7}{d_{\text{CNT}}} \quad (1.21)$$

where $v_F = 8 \times 10^5 \text{ m s}^{-1}$ is the Fermi velocity, E_g is expressed in eV, and d_{CNT} is expressed in nm.

1.3.2 Carbon Nanotube Field-Effect Transistors

General Operation

The modulation of a CNT channel by a gate-induced field was realised in the 1998 by the group of Cees Dekkar. These early FETs were fabricated by using doped silicon substrate as a gate to modulate the energy bands, separated from the nanotube by thermally-grown SiO_2 oxide acting as dielectric. For contacting the nanotube, metal contacts were employed.

Since then, a great variety of carbon nanotube field-effect transistor (CNTFET) structures and geometries have been demonstrated, and can be essentially classified in devices with or without a spacer. where the spacer length L_{spr} is the distance

between the gated channel and the source/drain metal contacts. Given that the nanotubes are intrinsically semiconducting, in spacer-less devices charge carriers are not induced in the channel by the gate-field but are injected from the contacts: the role of the gate is to lower the thermal barrier in the channel, thus allowing a tunnelling through the Schottky barrier (SB) at the source. In this scenario, a critical role for the CNTFET performance is played by the metal work function ϕ^M .

CNTFETs with spacers, on the other hand, are less limited by the injection at the junction source–CNT, as they are regulated by the modulating effect played by in the channel as in conventional. Since the carrier barrier injection at the drain is large, device with spacers have been proved to be more resistant to ambipolar conduction. In Table 1.2 an overview of gated-spacer (top and bottom) devices is reported.

Contacts

One of the principal issues when study the interface metal–CNT is the role played by carbon and metal atoms at the interface. Solid-state semiconductors have dangling bonds at the surface, leading to Fermi level pinning, meaning that the variation of the band bending is independent of the work function of the metal, even when this is large. Since the ideal structure of CNTs has no open bonds, no pinning of the metal Fermi level to mid-gap states occurs, thus legitimating the study of different work functions for maximising performance and tuning polarity. Many metals have been studied for contacting CNTs. Interestingly, it has been demonstrated that Pd ($\phi^{\text{Pd}} = 5.22 - -5.60$) builds an almost ideal *p*-channel contact to the valence band of CNTs. Despite having a larger workfunction ($\phi^{\text{Pt}} \approx 5.65$ eV) than Pd, Pt provides very poor contacts due to the physical wetting of the metal on the nanotube.

<i>Device type</i>	$I_{\text{on}}/I_{\text{off}}$	SS (mV/dec)	g_{mt} ($\mu\text{S cm}^{-1}$)	L_{ch} (nm)	f_T [f_{max}] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[37]
	10^5	~ 85	40	15	—	[38]
	10^5	~ 85	20	20	—	[38]
	10^5	60	20	200	—	[39]
	10^5	~ 63	—	200	—	[40]
	10^4	80	50	240	—	[41]
	—	—	143	800	[1]	[42]
	10^4	—	—	1,500	1.5 – 3.0	[43]
	—	70	$12^\dagger$	$2,000^1$	—	[44]
	10^8	—	—	$< 10,000^2$	—	[45]
	—	—	- 500	—	30	[46]
Top gate	10^6	—	20^2	80	—	[47]
	2	—	~ 50	100	25	[48]
	50	—	310	100	~ 70 [80]	[49]
	—	—	0.3	210	—	[50]
	10^5	—	33.5	85,000	—	[51]

Table 1.2: Figures of merit for a few CNFET reported in literature. S : subthreshold swing; g_m : intrinsic transconductances; L_{ch} : channel length; μ : carriers mobility

† Normalized by the tube width.

1 Referred to the gated length of a nanotube.

2 For the n-type transistor, whereas 10 μS were measured for the p-type.

<i>Device type</i>	I_{on} (A)	I_{off} (A)	$I_{\text{on}}/I_{\text{off}}$	g_{mt} (μS)	SS (mV/dec)
Conventional	1.41×10^{-5}	4.06×10^{-10}	3.4×10^4	29	99.6
Partial S/D-doped	1.50×10^{-5}	2.33×10^{-8}	6.4×10^2	63.6	93.1
Schottky-Barrier	3.26×10^{-6}	2.31×10^{-8}	1.41×10^2	19.6	109.8
Tunnel	5.53×10^{-6}	1.16×10^{-8}	4.76×10^2	13	107.9

Table 1.3: Comparative simulation for the four major type of CNFETs as reported in [52]. These results are obtained by setting $V_{\text{sd}} = 0.5$ V.

***n*-Type CNTFETs**

In a *n*-type CNTFETs the injection of electrons is favoured over the holes. Given that $\phi^{\text{CNT}} \approx 4.7$ eV and with an average $E_g \approx 0.6$ eV, it is possible to configure *n*-type channel with nanotube by either doping the spacer region or by using a metal with lower work function. One of the main drawbacks of doping is the challenge to keep it under control: instability in air, low consistency, variability to temperature variations, are amongst the most frequent scenarios. The other issue in working with low work function metals as contacts is the high tendency to oxidize. A simple and effective workaround to this is to use high work function metals to cap the contact metals, passivating the contact surface tendency to destabilise in air. One example of high performance *n*-type device was obtained by using Y contacts, achieving an Sub-threshold swing (SS) of 73 mV/dec, mobility of $5100 \text{ cm V}^{-2} \text{ s}^{-1}$, and drain induced barrier lowering (DIBL) of 105 mV V^{-1} [53].

Scaling

While Si technology was for long time questioned whether it could work below 10-nm of channel length, recent impressive technological developments from the industry have proved not only its feasibility but also a widespread commercial application (currently modern laptop employ the 5-nm technology). Nevertheless, the interest in Si scalability has been always a crucial starting point for exploring different technologies and materials capable of overcoming short-channel effects (threshold voltage roll-off, off-state leakage current, channel length modulation are just a few of the problematics).

In carbon nanotubes transistors a crucial and important – but somehow overlooked difference – is the difference between the two critical lengths for scaling, which are the contact length (L_c) and the channel length (L_{ch}). Aggressive scaling for the channel length generally leads to lose control of the band bending through the gate: intuitively, by shrinking the channel length the electric field generated by the source contact will be felt more and more by the charge carriers. The argument here is that carbon nanotubes offer, however, a good enough platform for such

scaling given by the ultrathin hollow structure (recall an effective wall thickness about $d_{\text{CNT}} \approx 0.68 \text{ \AA}$). On the other hand, this could be a double edge-sword. The effective carrier mass in nanotubes is $m_{\text{CNT}}^* = (0.228 \pm 0.129)m_0$ [54], thus potentially leading to tunnelling current from source to drain: this effect can be appreciate by an increase in the SS and the OFF current.

One of the first few systematic study on channel length scaling effects was conducted by Franklin *et al.* in 2010 through a series of measurements on nanotubes with relatively small diameters ($1.1 \text{ nm} < d_{\text{CNT}} < 1.3 \text{ nm}$) by using local bottom gates covered with 2 nm effective oxide thickness (EOT) of HfO_2 [55]. A consistent trend was observed when the channel length was scaled down more than two orders of magnitude (from 3 μm to 15 nm), both in the values of V_{th} and SS ($-0.5 \pm 0.2 \text{ V}$ and 90 mV/dec, respectively). The authors reported also encouraging figures on the transconductance g_m .

In a standard MOSFET, the transconductance is the change in the drain current divided by the amount varied in the gate/source voltage V_{gs} when the drain/source voltage (V_{ds}) is kept constant. Typical values of g_m for a small-signal field effect transistor are between 1 and 30 $\text{mS}\mu\text{m}^{-1}$. In a n-channel (nMOS) FET with threshold voltage V_{th} , the drain current at a certain V_{gs} in saturation mode (so when $V_{\text{th}} > V_{\text{gs}} - V_{\text{sd}}$) is given by:

$$I_{\text{sd}} = \frac{1}{2}\mu_n C_{\text{ox}} \frac{W}{L} (V_{\text{gs}} - V_{\text{th}})^2, \quad (1.22)$$

where μ_n is the electron mobility, C_{ox} is the gate oxide capacitance, W and L are the gate width and length, respectively. It is possible to work out the value of g_m by using the Shichman–Hodges model and assuming that the threshold voltage V_{th} is a device constant, such that $V_{\text{ov}} = V_{\text{gs}} - V_{\text{th}}$:

$$g_m = \frac{\partial I_{\text{sd}}}{\partial V_{\text{gs}}} = \frac{2 I_{\text{sd}}}{V_{\text{ov}}}, \quad (1.23)$$

Here V_{ov} is the so called *overdrive voltage* and has the physical meaning of being the voltage above the threshold voltage of the MOSFET. It is useful in the designing

phase of a transistor and elegantly links the drain current with a voltage: in fact, by rearranging Eq. 1.22 and identify $k_n = \mu_n C_{\text{ox}} \frac{W}{L}$ one gets:

$$I_{\text{sd}} = \frac{1}{2} k_n V_{\text{ov}}^2 \implies V_{\text{ov}} = \sqrt{\frac{2I_{\text{sd}}}{k_n}} \quad (1.24)$$

The usefulness for this equation is that the relation expressed in terms of overdrive voltage rather than threshold voltage holds both in the n and in the p case. Some reference numbers: V_{ov} is typically set at about 70–200 mV for the 65 nm technology node (with $I_{\text{sd}} = 1.13 \text{ mA } \mu\text{m}^{-1}$ of width) with value of $g_{\text{m}} \approx 11 - 32 \text{ mS } \mu\text{m}^{-1}$. In their work, Franklin *et al.* achieved transconductance peak up to $40 \text{ } \mu\text{S } \mu\text{m}^{-1}$ for a CNTFET with 15 nm of channel length, corresponding to values of g_{m} between approximately $91 - 110 \text{ } \mu\text{S } \mu\text{m}^{-1}$, thus meeting the requirements for being compatible with the Si technology.

The first CNTFET in which L_{ch} was scaled below 10 nm came two years later, again by Franklin *et al.* [37]. By using a CNT with $d_{\text{CNT}} \approx 1.3 \text{ nm} \pm 0.2 \text{ nm}$ and $E_g \approx 0.62 \text{ eV} \pm 0.1 \text{ eV}$ contacted through local bottom gates, they manage to scale down the $L_{\text{ch}} = 9 \text{ nm}$. The EOT was reduced to $\approx 0.65 \text{ nm}$ to achieve a better electrostatic coupling with the nanotube.

The authors reported an increase on the minimum current as L_{ch} decreases from 320 nm to 9 nm, attributed to a back-injection of the carriers into the conduction band from the drain, while the on-current approached the ballistic value for $L_{\text{ch}} \leq 41 \text{ nm}$ [55]. Impressive switching behaviour, with SS about 94 mV/dec, was measured, thus far outperforming any silicon-based of that time. based on previous results [55], the authors claimed the possibility of lowering the SS and the off-current at minimal cost to I_{on} by reducing the diameter of the nanotubes.

Contact Length Scaling

MOSFET devices, when shrunk down to the nanoscale, need to consider not only L_{ch} but also the length of the nanotube with the source and drain, L_c . The first study to address the impact of contact scaling in CNTFETs revealed the L_c to be inversely dependent on the channel length. Initial theoretical attempts to describe

this trend using dipole theories were started following the results obtained by the group of Hongjie Dai, where they showed it is possible to recover a perfect ohmic contact – thus accessing the ballistic transport regime – even with a semiconducting nanotube, which were at the time notorious of suffering several Schottky barriers effects (tunnelling and thermoionic emission).

Further investigations using more sophisticated techniques such as Density Functional Theory (DFT) coupled with the non-equilibrium Green function (NEGF) were employed to consider not only different contact geometries, but mostly to clarify the role in the chemical bonds formed between the metal to carbon structures, or in some combination of metal-/carbon-dependent interfacial charging properties. In particular, Cuniberti and coworkers addressed the chemisorption problem of carbon nanotubes with different electrodic surfaces. Chemisorption consists of the creation of a chemical bond between the atom deposited on a substrate (adatom) and its surface. The bonding leads to changes in the local density of states that produce specific alteration of the surface properties. They considered metal surfaces (such as Ni and Ti) inducing a chemical reaction with the carbon structure (chemisorption), elements such as Pd, Pt, Cr and Sc which interact weakly with the nanotube band structure leaving it unchanged, and surfaces which interact strongly through metallization and/or hybridisation of the π orbitals of the nanotube (Cu, Au, and Rh).

Low-ohmic contact scaling trends were found when using weakly and intermediate interacting metallic surfaces, while they envisioned optimal contact length scaling with metals such as Cu and Rh. While Rh contacts would be unaffordable at large scale due to manufacturing costs, it is well known that Cu presents ill-defined native oxide issues when used under ambient conditions, causing scattering at their surfaces and grain boundaries, thus hampering good electrical contacts with external probes. Recently, it has been advanced graphene as interlayer between the metal contact and the nanotube: the idea would be to have a work function closer to the nanotube one in order to avoid the level pinning effects due to gap states and interface contaminations.

1.4 Graphene Nanoribbons

Amongst the wide variety of carbon alternatives explored in the last two decades, graphene nanoribbons (GNRs) have certainly acquired a lot of interests. Amongst the perks, chemical synthesis of GNRs is now cost effective, they have comparable electrical performance (small effective mass, high charge mobility, and high saturation velocity) compared with the CNTs, without the need to separate metallic from semiconducting GNRs, and they can be easily tailored through chemical synthesis [56, 57].

As in the case of CNTs, confining large-area graphene carries into narrow stripes leads to a quasi-1D system exhibiting a tunable bandgap determined not only by the ribbon width w_{GNR} , but also by the edge structure [58, 59]. Logic applications, as mentioned in Sec. 1.2, require two well-defined voltage states, which in turn they depend on the current flowing through the semiconducting channel; by considering E_g within the 300 – 500 meV range, the figure (place the bandgap figure here) gives an indication circa suitable ribbon widths, which would lie within the sub-5 nm regime [60].

band structure

Historically, a first classification of graphene nanoribbons was derived by grouping the edge structures into Armchair-GNRs (AGNRs) and Zigzag-GNRs (ZGNRs). Following the success of chemical routes in producing GNRS, various kind of edges have come to light, e.g., cove, fjord, chiral, and all these can be regarded as combination of AGNRs and ZGNRs Figure.

Initial bandgap estimations through DFT calculations reported values between 2.5 and 0.5 eV. The inclusion of electronic correlation, however, becomes crucial for ultra-narrow (< 5 nm) ribbon [58, 61] and more accurate estimates derived by the group of Steven Louie based on the GW approximation reported a wider range of energy gaps from 5.5 to 1 eV.

These calculations have shown that, as rule of thumb, AGNRs can be both semiconducting and metallic, while ZGNRs should be regarded as quasi-metallic, with

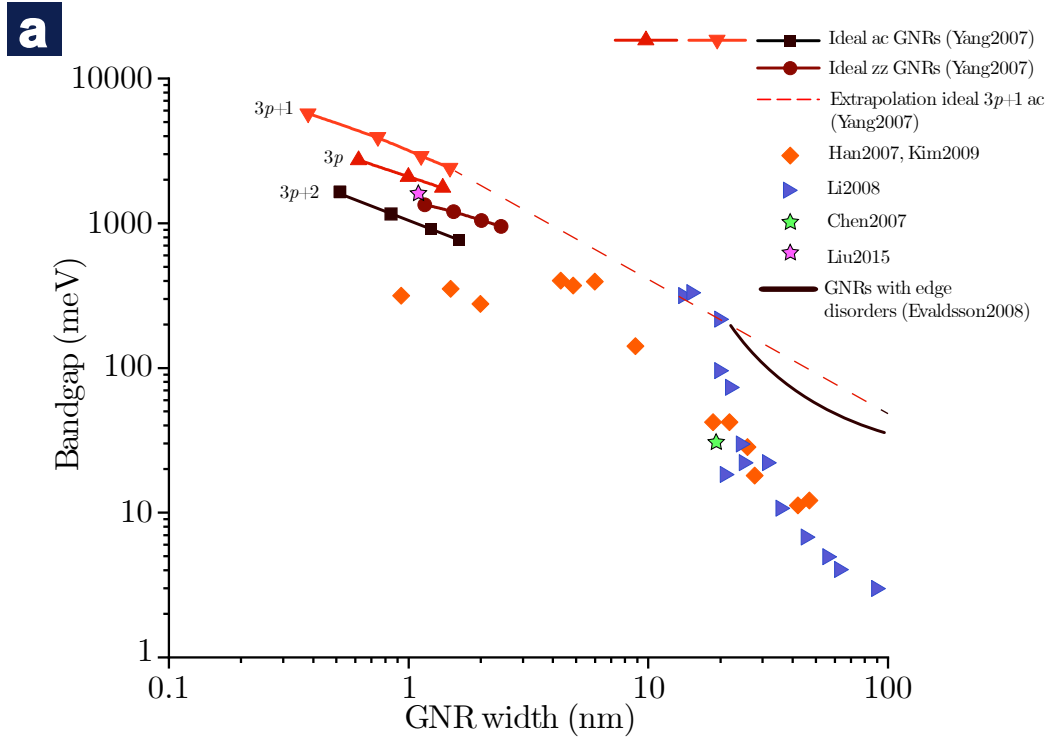


Figure 1.2: Majorana T-junction. Band gap versus nanoribbon width from experiments [59, 62–64] and theoretical calculations [65, 66]. The following abbreviation are used: ac for *armchair*, zz for *zigzag*. Figure adapted and extended from [25]

energy gaps lying within 0.1 - 0.7 eV. Indeed, for this kind of ribbon the application of an electric field in the transverse direction of induces a spin-polarisation of the edge states is believed to be responsible for semiconducting properties of ZGNRs. The simplest model for describing the energy levels for a graphene nanoribbon is the particle in the box model, where the energy should scale quadratically $E_g \propto 1/L^2$ with the ribbon extension. Experimentally, however, the band gap for AGNRs scales with the inverse of the width $E_g \propto 1/W$ [58]: by considering all the interatomic carbon-carbon distances the same, it is possible to work out an averaged formula which reports the band gap varying as $E_g = 0.8/W$, [67]. More accurate simulations have shown, however, that peripheral carbon—carbon bond lengths usually tend to be larger compared to the ribbon core ones [68]. It is predicted that the band gap will be similar to Ge or InN when the width is reduced to 2 and 3 nm, and will be similar to Si or GaAs when it is further reduced to 1-2 nm [69].

AGNR

For AGNRs, the bandgap varies significantly for different m values. AGNRs exhibit metallic behavior when $m = 3p + 2$, where p is an integer number, and they become semiconductors when $m = 3p$ or $3p + 1$. These relations have been obtained through *in silico* experiments, both using the tight-binding method and DFT. Figure 3 presents the isosurface plots of Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO) of AGNRs with $m=7,8,9$ respectively. Interestingly, the HOMO and LUMO orbitals are localized at the terminal of the ribbon along the length direction when $m=7(3p+1)$.

ZGNR

The band structure of ZGNRs is more interesting. Tight binding estimations have shown that if spin DOFs are excluded, the conduction band and valence band touches at the fermi level, forming a flat band. This flat band leads to a large localization of the density of states at the edges of ZGNRs, the so called *edge states*, which displays an exponential decay $\rho(\epsilon \sim \epsilon^{-r})$, with an exponent depending of the distance from centre of the ribbon; the weight of these states become negligible when the width w is the order of μm . The formation of band gap is observed when the spin is considered. In particular, the ground state of ZGNRs has parallel (ferromagnetic, $\uparrow\uparrow$, FM) spin alignment along one edge, whilst two edges have antiparallel (antiferromagnetic, $\downarrow\downarrow$, AFM) configuration. Given that the AFM state is semiconducting while the FM is metallic, it would theoretically be possible to change the metallic state of a zigzag ribbon by applying a magnetic field in the direction transverse to that of the ribbon's extension.

The scientific interest for graphene nanoribbons started in the physics community. Although this may appear irrelevant at the first glance, we will see that this actually had an impact over the last twenty years of progress in the field. The physics community was looking for a way to open a band gap in graphene. As mentioned in Sec. 1.1, large-area graphene is a semimetal, therefore unsuitable for complementary metal-oxide-semiconductor (CMOS) logic and problematic for RF.

The first class of methods for synthesising graphene nanoribbons are methods that involve cutting or producing rectangular shapes from the graphene lattice through macroscopic methods. For this reason, they are also known as *top-down* approaches. Although they have seen an improvements in recent years, common drawbacks for this class of methods are a random distribution in the widths, a non-atomical control of the edge structures, and non-negligible impurity concentrations left after the process.

Kim and co-workers synthesized GNRs by employing the electron-beam lithography and etching technique [70]. The widths of GNRs obtained by this technique were scattered over a broad range, and most of the edges were extremely rough. An 8 nm GNR FET showed a dramatically improved on/off ratio of about 160.

Nanocutting of graphene is another promising method of GNR synthesis. In 2008, Strachan *et al.* used thermally activated Fe nanoparticles as chemical scissors to cut graphene sheet along a specific crystallographic direction [70, 71]. Later, Ni and other catalysts were employed as well. The carbon-carbon bond breaking is driven by the combustion reaction $\text{Ni} + \text{C}_{\text{graphene}} + 2\text{H}_2 \uparrow \longrightarrow \text{Ni/Fe} + \text{CH}_4 \uparrow$ catalysed by the nanoparticles, thus producing hydrocarbon compounds. One of the reason that have discouraged these synthetic way is the random width distribution obtained: this is likely to be a consequence of the change in direction when nanoparticles encounter defects or edges.

edge states

Band gap opening and magnetic ordering are two main features of ZGNRs when the electron-electron correlations and the spin DOFs are considered. Steven Louie's group extended the computation also to chiral edges, elucidating the inherent role of spin-polarized magnetic edge states in these kind of ribbons. [72]. They employed a mean-field Hubbard model to calculate the band structure, assuming the magnitude for the on-site Coulomb repulsion being comparable to the hopping integral $U/t \sim 1$. The Hubbard hamiltonian using the second quantisation formalism resembles Eq. 1.11 and can be expressed as:

$$\begin{aligned} \mathcal{H}_{\text{MF}} = & -t \sum_{\langle ij \rangle, \sigma} (a_{\sigma,i}^\dagger a_{\sigma,j} + \text{H.c.}) \\ & + U \sum_i (n_{i\uparrow} \langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle n_{i\downarrow} - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle), \end{aligned} \quad (1.25)$$

except for the second term, which is given by the summation of spin-polarised density $n_{i\sigma} = a_{i\sigma}^\dagger a_{i\sigma}$ and their expectation values. The hamiltonian in Eq. 1.25 respects the electron-hole symmetry, meaning that the density of states is symmetric respect to the E_F . The solution of the model displays interesting features: two pairs of peaks split (Δ_0 and Δ_1) in the density of states are observed, one associated with the electronic band gap, and one associated with the maximum energy splitting. The authors claim that the first peak split arises due to an AFM correlation between opposite edges, while Δ_1 is a consequence of a ferromagnetic order due to strong localised state along a single edge. Within this description, it is legit to link the behaviour of chiral GNRs with the one of spin-polarised ZGNRs; thus the total energy difference between ferromagnetic and antiferromagnetic couplings should be in the order of tens of meV per edge atom and should decrease as the width of the ribbon increases, eventually becoming negligible when the ribbon width is significantly larger than the decay length of the spin polarized [73]. Both features are in fact recovered from the solution of Eq. 1.25.

GNRFET

First prototype of a field-effect transistor realised with very narrow graphene nanoribbon as channel material was realised in 2008 by the Dai's group, [74], using sonochemistry methods and global Si/SiO₂ backgate. In particular, with widths $w_{\text{GNR}} \approx 2.0(5) \text{ nm}$ and lengths of $l_{\text{GNR}} \approx 236 \text{ nm}$, they were able to obtained impressive transistor performance such as $I_{\text{on}} \approx 2000 \text{ } \mu\text{A } \mu\text{m}^{-1}$, $I_{\text{on}}/I_{\text{off}} \approx 10^6$, $SS = 210 \text{ mV/dec}$, $g_{\text{m}} = 900 \text{ } \mu\text{S } \mu\text{m}^{-1}$. This pioneer work sparked an exciting graphene nanoribbon field-effect transistors (GNRFETs) research (see table [table GNRFET](#)).

<i>Fabrication Approach</i>	Structural Parameter			Device Parameter	
	Edge Structure	Width (nm)	E_g (meV)	$I_{\text{on}}/I_{\text{off}}$	Mobility ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)
Top-Down	—	5	145	$47; 10^5$ (5.4 K) ⁷⁵	—
	—	5	489.4 ± 19.0	$> 10^4$ (290 K) ⁷⁶	—
	—	15	120 ± 23	— ⁷⁶	—
	Mixed	10	140	10^5 (4 K) ⁷⁷	—
	—	12	100	$10; 10^6$ (4 K) ⁷⁸	—
	—	sub-10	200	— ^{79†}	—
	—	10 – 20	10 – 15	$4.5; 10$ (20 K) ⁸⁰	—
	Closed	2.8	494	10^5	2443_{μ_h}
	Closed ⁸¹	9.9	$891/[\text{w}_{\text{gnr}} - 1.0]$	$< 10^2$	3776_{μ_h}
	—	100 – 400	18 – 1	$20 - 3.5^{82}$	—
Bottom-up	Cove	0.69 – 1.13	1880	$>^{\dagger} 10^{10}$	
	AGNR	0.95	2100	$^{\dagger}10^5$	
	Chevron	1.73	3100 ± 400	—	
	—	2	—	$> 10^6$	
	AGNR	0.63	1690 ± 100	$>^{\dagger} 10^{10}$	
	AGNR	0.95	1690 ± 70	—	
	AGNR	1.27	1130 ± 50	$^{\dagger}10^9$	
	AGNR	1.37	1600 ± 400	—	
	AGNR	1.58	1030 ± 40	$^{\dagger}10^8$	
	AGNR	5 – 9	≤ 400	$10^5 - -10^6$	
	ZGNR	6	1400	—	
Epitaxial	—	900	—	—	
	AGNR	2 – 40	> 500	—	
	—	40	—	$10; > 25$ (4 K)	

Table 1.4: Differences between the main semiconductors used in modern digital electronics and graphene.

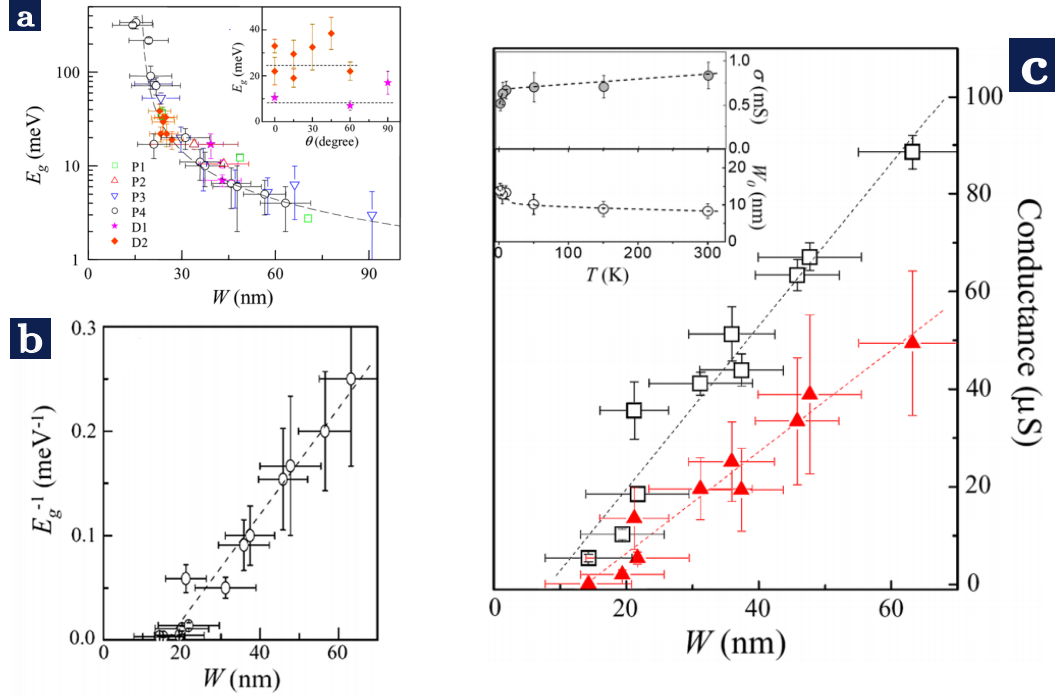


Figure 1.3: (a) E_g vs W for four (P1–P4) of the parallel devices and two (D1, D2) with varying orientation. The inset shows E_g vs relative angle θ for the device D1 and D2. (b) E_g^{-1} vs W . (c) Conductance vs width of parallel GNRs measured at a gate voltage $V - V_g = 50$ V. Squares are values measure at 300 K whereas triangle corresponds to 1.6 K. The insets show the conductivity (upper) and the inactive GNR width (lower) obtained from the slope and x intercept of the linear fit at varying temperatures. Adapted from [62].

2

Foundations

Contents

1.1	Motivation	1
1.2	Graphene	1
1.2.1	crystal structure	1
1.2.2	band dispersion	3
1.2.3	Graphene Transistors	5
1.3	Carbon-based Field Effect Transistor	12
1.3.1	crystal structure	12
1.3.2	Carbon Nanotube Field-Effect Transistors	13
1.4	Graphene Nanoribbons	20

2.1 Charge Transport In Zero-Dimensional Objects

quantum dots transport physics

2.2 Charge Transport in One-Dimensional Objects

nanowires transport physics

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

What are the necessary changes that one has to take into account when dealing also with other extensive variable such as a temperature gradient

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

3

Instrumentations

Contents

2.1	Charge Transport In Zero-Dimensional Objects	27
2.2	Charge Transport in One-Dimensional Objects	27
2.3	Thermoelectric Phenomena in Zero-Dimensional Structures	28
2.4	Density Functional Theory	28
2.5	Non-Equilibrium Green Function Theory	28

3.1 Room Temperature DC Measurements

Describe here the probe station setup]

3.2 Dipstick DC Measurements

Describe here the dipstick used for both liquid N₂ and liquid He

3.3 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

3.4 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it up

3.5 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

4

Fabrication

Contents

3.1	Room Temperature DC Measurements	29
3.2	Dipstick DC Measurements	29
3.3	Cryogenic DC Measurements	29
3.4	Cryogenic AC Measurements	30
3.5	Thermoelectric Measurements	30

4.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

4.1.1 Graphene Preparation and Wet Transferring

Initially the graphene was transferred on top of the substrate through the wet transfer method. The graphene was supplied by Graphene ain 5 x 5 cm2 rectangles, with a PMMA front-layer and a back-layer of partially removed graphene. Before transfer, the back-layer had to be fully removed via etching, and the remaining stack appropriately sized for the substrates for this work. The graphene arrived in a vacuum-sealed pack, which had to be transported to the cleanroom before

being opened and the following preparation steps initiated. Figure 6.2 depicts the main stages involved in preparing the graphene for transfe

4.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and lithographic patterning of graphene nanogap

4.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

4.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

4.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

4.3.3 Thermoelectric Device

Common and different features of the thermoelectric

4.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techniques How this has been qualitatevely addressed.

5

Room Temperature Graphene Nanoribbon Single-Electron Transistor

Contents

4.1	General Fabrication Procedure	31
4.1.1	Graphene Preparation and Wet Transferring	31
4.2	Graphene Nanogaps	32
4.3	Chip Design	32
4.3.1	2T Design	32
4.3.2	3T Design	32
4.3.3	Thermoelectric Device	32
4.4	Deposition Process and Related Phenomna	32

5.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)
Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

5.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

5.4 Summary

6

Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

Contents

5.1	Introduction	33
5.2	Sample Preparation and Characterisation	33
5.3	Electrical Transport Measurements	34
5.4	Summary	34

6.1 Introduction

A brief intro on how why it is important to study this

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)
Explained why the scarcity of characterisation checks i s intrinsically scarce in
molecular electronics.

6.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

6.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

6.5 Summary

7

Thermocurrent Spectroscopy of covalently Functionalized Graphene Nanoribbons

Contents

6.1	Introduction	35
6.2	Sample Preparation and Characterisation	35
6.3	<i>Ab-initio</i> Experiments	35
6.4	Electrical Transport Measurements	36
6.5	Summary	36

7.1 Introduction

A brief intro on how why it is important to study this what has been done.

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

7.4 Summary

8

Transport Spectroscopy of Biaryl-linked Dy–Tb Porphyrin

Contents

7.1	Introduction	37
7.2	Sample Preparation and Characterisation	37
7.3	Electrical Transport Measurements	38
7.4	Summary	38

8.1 Introduction

A brief intro on how why it is important to study this

8.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks i s intrinsically scarce in
molecular electronics.

8.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

8.4 Summary

9

Conclusions and Outlook

Contents

8.1	Introduction	39
8.2	Sample Preparation and Characterisation	39
8.3	Electrical Transport Measurements	40
8.4	Summary	40

9.1 Conclusions

Here the final conclusions

9.2 Outlook

Appendices

Cor animalium, fundamentum est vitæ, princeps omnium, Microcosmi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [83]



Review of Cardiac Physiology and Electrophysiology

Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec accumsan cursus neque. Pellentesque eget tempor turpis, quis malesuada dui. Proin egestas, sapien sit amet feugiat vulputate, nunc nibh mollis nunc, nec auctor turpis purus sed metus. Aenean consequat leo congue volutpat euismod. Vestibulum et vulputate

nisl, at ultrices ligula. Cras pulvinar lacinia ipsum at bibendum. In ac augue ut ante mollis molestie in a arcu.

Etiam vitae quam sollicitudin, luctus tortor eu, efficitur nunc. Vestibulum maximus, ante quis consequat sagittis, augue velit luctus odio, in scelerisque arcu magna id diam. Proin et mauris congue magna auctor pretium id sit amet felis. Maecenas sit amet lorem ipsum. Proin a risus diam. Integer tempus eget est condimentum faucibus. Suspendisse sem metus, consequat vel ante eget, porttitor maximus dui. Nunc dapibus tincidunt enim, non aliquam diam vehicula sed. Proin vel felis ut quam porta tempor. Vestibulum elit mi, dictum eget augue non, volutpat imperdiet eros. Praesent ac egestas neque, et vehicula felis.

Pellentesque malesuada volutpat justo, id eleifend leo pharetra at. Pellentesque feugiat rutrum lobortis. Curabitur hendrerit erat porta massa tincidunt rutrum. Donec tincidunt facilisis luctus. Aliquam dapibus sodales consectetur. Suspendisse lacinia, ipsum sit amet elementum fermentum, nulla urna mattis erat, eu porta metus ipsum vel purus. Fusce eget sem nisl. Pellentesque dapibus, urna vitae tristique aliquam, purus leo gravida nunc, id faucibus ipsum magna aliquet ligula. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin sem lacus, rutrum eget efficitur sed, aliquam vel augue. Aliquam ut eros vitae sem cursus ultrices ut ornare urna. Nullam tempor porta enim, in pellentesque arcu commodo quis. Interdum et malesuada fames ac ante ipsum primis in faucibus. Curabitur maximus orci purus, ut molestie turpis pellentesque ut.

Donec lacinia tristique ultricies. Proin dignissim risus ut dolor pulvinar mollis. Proin ac turpis vitae nibh finibus ullamcorper viverra quis felis. Mauris pellentesque neque diam, id feugiat diam vestibulum vitae. In suscipit dui eu libero ultrices, et sagittis nunc blandit. Aliquam at aliquet ex. Nullam molestie pulvinar ex vitae interdum. Praesent purus nunc, gravida id est consectetur, convallis elementum nulla. Praesent ex dolor, maximus eu facilisis at, viverra eget nulla. Donec ullamcorper ante nisi. Sed volutpat diam eros. Nullam egestas neque non tortor aliquet, sed pretium velit tincidunt. Aenean condimentum, est ac vestibulum mattis, quam

augue congue augue, mattis ultrices nibh libero non ante. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Aenean volutpat eros tortor, non convallis sapien blandit et. Maecenas faucibus nulla a magna posuere commodo. Nullam laoreet ante a turpis laoreet malesuada. Phasellus in varius sem. Vestibulum sagittis nibh sed tincidunt blandit. Donec aliquam accumsan odio sit amet lacinia. Integer in tellus diam. Vivamus varius massa leo, vitae ullamcorper metus pulvinar sed. Maecenas nec lorem ornare, elementum est quis, gravida massa. Suspendisse volutpat odio ex, ac ultrices leo ultrices vel. Sed sed convallis ipsum. Pellentesque euismod a nulla sed rhoncus. Sed vehicula urna vitae mi aliquet, non sodales lacus ullamcorper. Duis mattis justo turpis, id tempus est tempus eu. Curabitur vitae hendrerit ligula.

Curabitur non pretium enim, in commodo ligula. Etiam commodo eget ligula a lacinia. Vestibulum laoreet ante tellus, vel congue sapien ornare in. Donec venenatis cursus velit vitae pulvinar. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Suspendisse in metus lectus. Pellentesque gravida dolor eget finibus imperdiet. Duis id molestie tortor. Mauris laoreet faucibus facilisis. Aliquam vitae dictum massa, sit amet dignissim lacus.

Fusce eleifend tellus id ex consequat maximus. Donec ultrices ex ut turpis ornare, non molestie mi placerat. Nulla sit amet auctor nunc, sit amet euismod elit. Phasellus risus tellus, condimentum a metus et, venenatis tristique urna. Cras mattis felis eget ipsum fermentum egestas. Ut augue odio, venenatis id convallis vel, congue quis augue. Maecenas sed maximus est, posuere aliquet tortor. Ut condimentum egestas nisi eu porttitor. Ut mi turpis, posuere id lorem vel, elementum tempor arcu.

Morbi nisl arcu, venenatis non metus ac, ullamcorper scelerisque justo. Nulla et accumsan lorem. Mauris aliquet dui sit amet libero aliquet, in ornare metus porttitor. Integer ultricies urna eu consequat ultrices. Maecenas a justo id purus ultricies posuere sed et quam. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Sed eleifend risus quis aliquet gravida. Nullam ac erat porta est bibendum dictum in a dolor. Nam eget turpis viverra, vulputate

lectus eget, mattis ligula. Nam at tellus eget dui lobortis sodales et ut augue. In vestibulum diam eget mi cursus, ut tincidunt nulla pellentesque.

Aliquam erat volutpat. Sed ultrices massa id ex mattis bibendum. Nunc augue magna, ornare at aliquet gravida, vehicula sed lorem. Quisque lobortis ipsum eu posuere eleifend. Duis bibendum cursus viverra. Nam venenatis elit leo, vitae feugiat quam aliquet sed. Cras velit est, tempus ac lorem sed, pharetra lobortis ipsum. Donec suscipit gravida interdum. Nunc non finibus est. Nullam turpis elit, tempus non ante.

A.2 Mechanical Cycle

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean tellus est, suscipit sed facilisis quis, malesuada at ipsum. Nam tristique urna quis quam iaculis, et mattis orci pretium. Praesent euismod elit vel metus commodo ultrices. Vestibulum et tincidunt ex, in molestie ex. Donec ullamcorper sollicitudin accumsan. Etiam ac leo turpis. Duis a tortor felis. Nullam sollicitudin eu purus ac hendrerit. Nam hendrerit ligula libero, eget finibus orci bibendum a. Aenean ut ipsum magna.

Ut viverra, sapien sed accumsan blandit, nisi sem tempus tellus, at suscipit magna erat ornare nunc. Proin lacinia, nisi ut rutrum malesuada, nibh quam pellentesque nunc, sit amet consectetur purus felis ac tortor. Suspendisse lacinia ipsum eu sapien pellentesque mattis. Mauris ipsum nunc, placerat non diam vel, efficitur laoreet nunc. Sed lobortis, ipsum eget gravida facilisis, sem nulla viverra mi, in placerat eros sem viverra lacus. Aliquam porta aliquet diam vel commodo. Nulla facilisi. Duis erat libero, lobortis vel hendrerit vitae, sagittis id dui. Nulla pretium eros nec quam tincidunt, vel luctus mi aliquam. Integer imperdiet purus in est tristique venenatis. Ut pellentesque, nunc vitae iaculis ultricies, urna turpis dignissim risus, a laoreet felis magna nec erat.

Quisque sollicitudin faucibus ligula, et egestas nibh dictum sit amet. Proin eu mi a lectus congue pretium eu quis arcu. Suspendisse vehicula libero eu ipsum aliquam, vel elementum nibh mattis. Sed sed sapien vitae turpis tristique pulvinar a ut metus. Etiam semper gravida est, mollis gravida est porta ac. Proin eget tincidunt

erat. Maecenas ultrices erat eget purus ultricies, ut lacinia arcu dictum. Nam et nisi sit amet ex congue mattis vel eget lorem. Aliquam erat volutpat. Pellentesque porttitor nibh vitae elementum consectetur. Aenean et est lobortis, congue sapien non, ullamcorper sapien. Ut facilisis sem non dapibus vehicula.

Mauris euismod odio dolor, sit amet gravida mauris placerat et. Curabitur nec dolor non nibh molestie lobortis dignissim non ante. Nullam rutrum lobortis ultrices. Aenean ex erat, elementum sed maximus id, posuere id quam. Proin rutrum ex elit, pretium aliquam risus finibus at. Aenean egestas orci velit, sed aliquet sapien condimentum a. Duis consequat, arcu eu viverra venenatis, dolor lorem gravida lectus, non aliquet nisi sem at augue. Donec laoreet blandit luctus. Aenean vehicula nisl vel faucibus luctus. Sed ut semper velit, vitae laoreet magna. Sed at interdum magna.

Sed iaculis faucibus odio, eu aliquam purus efficitur vel. Cras at nulla ac enim congue varius ut et nulla. Integer blandit mattis augue.

A.3 Electrical Cycle

Lorem ipsum dolor sit amet, consectetur adipiscing elit. In faucibus condimentum rhoncus. Ut dictum nisl id risus gravida lobortis. Sed vehicula mollis tellus ut varius. Fusce eget egestas dui, et commodo dui. Proin sollicitudin interdum tempus. Nullam in elit a enim fringilla bibendum. Vestibulum sodales pellentesque condimentum. Nulla facilisi. Nunc et dolor in nulla eleifend dictum at vel ligula. Aliquam ut velit non elit ullamcorper porta ac et ex. Fusce ornare magna non nunc vestibulum, eget molestie quam dictum. In interdum aliquam odio, in posuere tellus convallis quis. Curabitur non diam elit. Proin vulputate orci diam, a tincidunt ante luctus eu. Ut a viverra ligula. Curabitur pulvinar tempus tellus eget suscipit.

Aliquam posuere massa at ante dapibus congue. Curabitur ullamcorper tortor eget consectetur aliquet. Mauris tempor magna id mauris fringilla, a varius erat blandit. Nam eleifend ullamcorper placerat. Phasellus augue tortor, volutpat bibendum lorem nec, fringilla volutpat nisl. Mauris cursus urna metus, vel eleifend orci iaculis ut. Sed sit amet scelerisque massa, quis consequat dui. Donec

semper sem dui, ac placerat velit egestas vel. Nulla facilisi. Quisque tellus eros, sagittis malesuada augue ut, faucibus dictum nulla. Vestibulum non dapibus erat, ut consequat libero. Ut turpis mi, dapibus commodo libero lobortis, maximus vestibulum lectus. Vestibulum sit amet sapien dapibus, tincidunt leo in, suscipit arcu. Sed in erat bibendum, laoreet eros eu, pellentesque justo. Nulla sodales purus neque, eget maximus ipsum consequat at. Maecenas a nisl sagittis, tempus ipsum sed, dictum mauris.

Suspendisse posuere odio lacus, at auctor tortor vehicula sed. Phasellus suscipit ornare enim vitae placerat. Sed viverra purus vel sapien tempor, quis iaculis erat laoreet. Aenean vel nunc vestibulum, ornare nunc ac, mollis urna. Aenean ultrices felis ipsum, ac semper est ullamcorper in. Donec in justo varius, egestas tortor ut, venenatis augue. Duis mattis, ligula quis lacinia fringilla, tellus neque accumsan ipsum, vitae tempor metus elit vel nibh. Curabitur porttitor urna nec sapien tempor, et porttitor velit malesuada.

Suspendisse aliquam nisl quis placerat vulputate. Proin dapibus ipsum ac ante sagittis, volutpat auctor sem dapibus. Nam in facilisis odio. Integer ante mauris, eleifend et pulvinar in, venenatis quis ligula. Phasellus posuere sollicitudin tortor eget euismod. Maecenas mollis tortor eget justo vulputate sagittis. Etiam hendrerit massa quis ex molestie sodales. Quisque facilisis erat lacus, id convallis sem suscipit bibendum. Integer dui urna, pharetra sed porta sed, bibendum ut odio. Donec placerat at lectus egestas consequat. Sed id rhoncus est, vitae vulputate sapien. Fusce tempus quam lorem, id ornare turpis sodales sed. Integer aliquet urna eget condimentum consequat. Vestibulum quis dui vel ligula posuere luctus id nec turpis.

Nam vitae placerat lacus. Mauris scelerisque interdum volutpat. Nunc aliquet tristique enim, sit amet molestie felis ullamcorper vitae. Nullam sollicitudin orci orci, in condimentum tellus consectetur in. Nam id justo justo. Fusce eget finibus est. Proin id tortor nec quam cursus vehicula. Aliquam ultrices eros eros, a tincidunt elit eleifend auctor.

Nullam consectetur dapibus ligula sit amet efficitur. Nunc non posuere sapien. Vivamus dui nisl, aliquam id ipsum non, pulvinar ornare neque. Nunc rhoncus

pretium congue. Fusce id laoreet enim. Cras sed massa in eros bibendum auctor in nec sem. Nam commodo, velit id porta consequat, mi arcu gravida lorem, ut aliquam elit ante quis dui. Quisque in massa sed nibh blandit dictum.

Vestibulum molestie consectetur porttitor. Donec tincidunt vel orci at pharetra. Nullam id felis sit amet nulla tempus lacinia. Integer egestas ullamcorper massa, ut ultricies diam congue sit amet. Cras sit amet velit at nibh vehicula finibus a et lorem. Cras odio metus, venenatis ut ultrices non, ornare ac orci. Morbi et nulla dui. Mauris dictum molestie nibh, eu efficitur lorem accumsan quis.

A.4 Cellular Electromechanical Coupling

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam vitae consectetur metus, ac maximus ex. Quisque vitae ex eu lectus ultricies consequat vel non lorem. Etiam odio ipsum, tempus ut lobortis in, molestie ac leo. Vivamus mollis feugiat bibendum. Vestibulum eget venenatis quam. Aenean faucibus, massa sed ullamcorper porta, arcu nunc iaculis velit, quis consectetur purus neque placerat nibh. Vestibulum elit nunc, dignissim vulputate venenatis et, sodales non massa. Proin leo ligula, vehicula eu aliquam varius, posuere a dolor. Donec iaculis auctor neque, sit amet gravida libero porta vel. Vivamus consequat elementum lacus, at bibendum mauris egestas nec. Fusce fermentum diam eu dolor ornare, vitae vestibulum leo interdum. Morbi luctus libero quis dictum laoreet. Etiam semper porta ante, vel ullamcorper enim sodales quis.

Nullam eu nisi faucibus, fermentum ex auctor, tempor arcu. Phasellus condimentum erat mi, condimentum malesuada ligula congue venenatis. Nullam gravida imperdiet urna quis cursus. Ut tempus nec purus eget posuere. Cras non nulla sit amet justo aliquet pellentesque nec sed eros. Nam aliquam nisl urna, in placerat magna gravida venenatis. Donec interdum vel magna ullamcorper molestie. Nunc felis neque, rhoncus fringilla faucibus sit amet, ultrices sed magna. Maecenas malesuada hendrerit diam in ultrices. Nam libero urna, volutpat ut auctor eget, interdum sed odio. Vestibulum suscipit mauris nec augue ornare, ut eleifend nulla gravida. Proin imperdiet, mauris quis consectetur porta, leo dui convallis leo,

id lobortis massa diam eu libero. Aenean hendrerit vel ante aliquam venenatis. Pellentesque bibendum pretium odio, ut sagittis lectus feugiat a. Donec porttitor vulputate lacus.

Nunc volutpat efficitur lacus in aliquet. Nullam non iaculis diam, at ultrices diam. Proin vehicula vulputate cursus. Morbi tempus sapien id urna lobortis interdum. Maecenas elementum sagittis elementum. Donec at sodales velit, a posuere tortor. Nulla id hendrerit tortor. Sed semper velit in magna sagittis pulvinar. Nulla nec arcu molestie, ultricies sapien sit amet, sollicitudin nisi. Donec nisi massa, suscipit ut dignissim quis, lacinia id leo.

Suspendisse ut mi metus. Morbi tincidunt ligula in porttitor consectetur. Integer eu urna urna. Suspendisse potenti. Mauris sit amet felis eu diam auctor ullamcorper. Morbi in porta nisi. Nam ante tortor, venenatis vitae tempor sed, sagittis vitae velit. In semper orci sit amet nisi ullamcorper varius. Aenean dignissim ultrices imperdiet. Maecenas lacinia enim id neque porttitor iaculis. Curabitur laoreet ante ut urna dignissim, id sollicitudin metus consectetur. Aenean massa ipsum, auctor vel ante vel, blandit dignissim libero. Fusce interdum ac magna et interdum.

*The first kind of intellectual and artistic personality
belongs to the hedgehogs, the second to the foxes ...*

— Sir Isaiah Berlin [84]

References

- [1] A. H. Castro Neto et al. “The electronic properties of graphene”. In: *Rev. Mod. Phys.* 81.1 (Jan. 2009), pp. 109–162. URL: <http://link.aps.org/doi/10.1103/RevModPhys.81.109>.
- [2] Cristina Bena and Steven A Kivelson. “Quasiparticle scattering and local density of states in graphite”. In: *Physical Review B* 72.12 (2005), p. 125432.
- [3] Frank Schwierz, Hei Wong, and Juin J Liou. *Nanometer CMOS*. Pan Stanford Publishing, 2010.
- [4] Kostya S Novoselov et al. “Electric field effect in atomically thin carbon films”. In: *science* 306.5696 (2004), pp. 666–669.
- [5] Kirill I Bolotin et al. “Ultrahigh electron mobility in suspended graphene”. In: *Solid state communications* 146.9-10 (2008), pp. 351–355.
- [6] Lei Wang et al. “One-dimensional electrical contact to a two-dimensional material”. In: *Science* 342.6158 (2013), pp. 614–617.
- [7] Cory R Dean et al. “Boron nitride substrates for high-quality graphene electronics”. In: *Nature nanotechnology* 5.10 (2010), pp. 722–726.
- [8] Kinam Kim et al. “A role for graphene in silicon-based semiconductor devices”. In: *Nature* 479.7373 (2011), pp. 338–344.
- [9] Frank Schwierz. “Graphene transistors”. In: *Nature nanotechnology* 5.7 (2010), p. 487.
- [10] Dharmendar Reddy et al. “Graphene field-effect transistors”. In: *Journal of Physics D: Applied Physics* 44.31 (2011), p. 313001.
- [11] Haomin Wang et al. “Hysteresis of electronic transport in graphene transistors”. In: *ACS nano* 4.12 (2010), pp. 7221–7228.
- [12] P Joshi et al. “Intrinsic doping and gate hysteresis in graphene field effect devices fabricated on SiO₂ substrates”. In: *Journal of Physics: Condensed Matter* 22.33 (2010), p. 334214.
- [13] Max C Lemme et al. “A graphene field-effect device”. In: *IEEE Electron Device Letters* 28.4 (2007), pp. 282–284.
- [14] JS Moon et al. “Epitaxial-graphene RF field-effect transistors on Si-face 6H-SiC substrates”. In: *IEEE Electron Device Letters* 30.6 (2009), pp. 650–652.
- [15] Antonio Di Bartolomeo et al. “Charge transfer and partial pinning at the contacts as the origin of a double dip in the transfer characteristics of graphene-based field-effect transistors”. In: *Nanotechnology* 22.27 (2011), p. 275702.
- [16] Inanc Meric et al. “Current saturation in zero-bandgap, top-gated graphene field-effect transistors”. In: *Nature nanotechnology* 3.11 (2008), p. 654.

- [17] Jakub Kedzierski et al. “Epitaxial graphene transistors on SiC substrates”. In: *IEEE Transactions on Electron Devices* 55.8 (2008), pp. 2078–2085.
- [18] Y-M Lin et al. “100-GHz transistors from wafer-scale epitaxial graphene”. In: *Science* 327.5966 (2010), pp. 662–662.
- [19] L Britnell et al. “Field-effect tunneling transistor based on vertical graphene heterostructures”. In: *Science* 335.6071 (2012), pp. 947–950.
- [20] Thanasis Georgiou et al. “Vertical field-effect transistor based on graphene–WS 2 heterostructures for flexible and transparent electronics”. In: *Nature nanotechnology* 8.2 (2013), p. 100.
- [21] Li-Hong Liu et al. “A simple and scalable route to wafer-size patterned graphene”. In: *Journal of materials chemistry* 20.24 (2010), pp. 5041–5046.
- [22] Zhengzong Sun, Dustin K James, and James M Tour. “Graphene chemistry: synthesis and manipulation”. In: *The Journal of Physical Chemistry Letters* 2.19 (2011), pp. 2425–2432.
- [23] Chuc Van Nguyen et al. “Effect of glass surface treatments on the deposition of highly transparent reduced graphene oxide films by dropcasting method”. In: *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 498 (2016), pp. 231–238.
- [24] Inanc Meric et al. “RF performance of top-gated, zero-bandgap graphene field-effect transistors”. In: *2008 IEEE International Electron Devices Meeting*. IEEE. 2008, pp. 1–4.
- [25] Frank Schwierz. “Graphene transistors: status, prospects, and problems”. In: *Proceedings of the IEEE* 101.7 (2013), pp. 1567–1584.
- [26] Yu-Ming Lin et al. “Development of graphene FETs for high frequency electronics”. In: *2009 IEEE International Electron Devices Meeting (IEDM)*. IEEE. 2009, pp. 1–4.
- [27] Lei Liao et al. “High-speed graphene transistors with a self-aligned nanowire gate”. In: *Nature* 467.7313 (2010), p. 305.
- [28] Simon M Sze, Yiming Li, and Kwok K Ng. *Physics of semiconductor devices*. John wiley & sons, 2021.
- [29] Rui Cheng et al. “High-frequency self-aligned graphene transistors with transferred gate stacks”. In: *Proceedings of the National Academy of Sciences* 109.29 (2012), pp. 11588–11592.
- [30] Yanqing Wu et al. “State-of-the-art graphene high-frequency electronics”. In: *Nano letters* 12.6 (2012), pp. 3062–3067.
- [31] S Sonde et al. “Electrical properties of the graphene/4 H-SiC (0001) interface probed by scanning current spectroscopy”. In: *Physical Review B* 80.24 (2009), p. 241406.
- [32] K.L. Tan et al. “94-GHz 0.1- μ m T-gate low-noise pseudomorphic InGaAs HEMTs”. In: *IEEE Electron Device Letters* 11.12 (1990), pp. 585–587.
- [33] I. Post et al. “A 65nm CMOS SOC Technology Featuring Strained Silicon Transistors for RF Applications”. In: *2006 International Electron Devices Meeting*. 2006, pp. 1–3.

- [34] R Lai et al. “Sub 50 nm InP HEMT device with Fmax greater than 1 THz”. In: *2007 IEEE International Electron Devices Meeting*. IEEE. 2007, pp. 609–611.
- [35] Mildred S Dresselhaus et al. “Carbon nanotubes”. In: *The physics of fullerene-based and fullerene-related materials*. Springer, 2000, pp. 331–379.
- [36] Edward A Laird et al. “Quantum transport in carbon nanotubes”. In: *Reviews of Modern Physics* 87.3 (2015), p. 703.
- [37] Aaron D Franklin et al. “Sub-10 nm carbon nanotube transistor”. In: *Nano letters* 12.2 (2012), pp. 758–762.
- [38] Aaron D Franklin and Zhihong Chen. “Length scaling of carbon nanotube transistors”. In: *Nature nanotechnology* 5.12 (2010), p. 858.
- [39] Ralf Thomas Weitz et al. “High-performance carbon nanotube field effect transistors with a thin gate dielectric based on a self-assembled monolayer”. In: *Nano letters* 7.1 (2007), pp. 22–27.
- [40] Yu-Ming Lin et al. “High-performance carbon nanotube field-effect transistor with tunable polarities”. In: *IEEE Transactions on Nanotechnology* 4.5 (2005), pp. 481–489.
- [41] Ali Javey et al. “High-field quasiballistic transport in short carbon nanotubes”. In: *Physical Review Letters* 92.10 (2004), p. 106804.
- [42] N Chimot et al. “Gigahertz frequency flexible carbon nanotube transistors”. In: *Applied Physics Letters* 91.15 (2007), p. 153111.
- [43] Michael Schröter et al. “CNTFET-based RF electronics—State-of-the-art and future prospects”. In: *2016 IEEE 16th Topical Meeting on Silicon Monolithic Integrated Circuits in RF Systems (SiRF)*. IEEE. 2016, pp. 97–100.
- [44] Ali Javey et al. “High- κ dielectrics for advanced carbon-nanotube transistors and logic gates”. In: *Nature materials* 1.4 (2002), p. 241.
- [45] Vladimir Derenskyi et al. “Carbon nanotube network ambipolar field-effect transistors with 108 on/off ratio”. In: *Advanced Materials* 26.34 (2014), pp. 5969–5975.
- [46] A Le Louarn et al. “Intrinsic current gain cutoff frequency of 30 GHz with carbon nanotube transistors”. In: *Applied physics letters* 90.23 (2007), p. 233108.
- [47] Ali Javey et al. “High performance n-type carbon nanotube field-effect transistors with chemically doped contacts”. In: *Nano letters* 5.2 (2005), pp. 345–348.
- [48] Yuchi Che et al. “T-gate aligned nanotube radio frequency transistors and circuits with superior performance”. In: *ACS nano* 7.5 (2013), pp. 4343–4350.
- [49] Yu Cao et al. “Radio frequency transistors using aligned semiconducting carbon nanotubes with current-gain cutoff frequency and maximum oscillation frequency simultaneously greater than 70 GHz”. In: *ACS nano* 10.7 (2016), pp. 6782–6790.
- [50] Fumiyuki Nihey et al. “A top-gate carbon-nanotube field-effect transistor with a titanium-dioxide insulator”. In: *Japanese journal of applied physics* 41.10A (2002), p. L1049.
- [51] Pak Heng Lau et al. “Fully printed, high performance carbon nanotube thin-film transistors on flexible substrates”. In: *Nano letters* 13.8 (2013), pp. 3864–3869.

- [52] Amandeep Singh, Mamta Khosla, and Balwinder Raj. “Comparative analysis of carbon nanotube field effect transistors”. In: *2015 IEEE 4th Global Conference on Consumer Electronics (GCCE)*. IEEE. 2015, pp. 552–555.
- [53] Li Ding et al. “Y-contacted high-performance n-type single-walled carbon nanotube field-effect transistors: scaling and comparison with Sc-contacted devices”. In: *Nano letters* 9.12 (2009), pp. 4209–4214.
- [54] Jose M Marulanda and Ashok Srivastava. “Carrier density and effective mass calculations in carbon nanotubes”. In: *physica status solidi (b)* 245.11 (2008), pp. 2558–2562.
- [55] Aaron D Franklin and Zhihong Chen. “Length scaling of carbon nanotube transistors”. In: *Nature nanotechnology* 5.12 (2010), p. 858.
- [56] Jinming Cai et al. “Atomically precise bottom-up fabrication of graphene nanoribbons”. In: *Nature* 466.7305 (2010), p. 470.
- [57] Matthias Georg Schwab et al. “Structurally defined graphene nanoribbons with high lateral extension”. In: *Journal of the American Chemical Society* 134.44 (2012), pp. 18169–18172.
- [58] Young-Woo Son, Marvin L Cohen, and Steven G Louie. “Energy gaps in graphene nanoribbons”. In: *Physical review letters* 97.21 (2006), p. 216803.
- [59] Xiaolin Li et al. “Chemically derived, ultrasmooth graphene nanoribbon semiconductors”. In: *science* 319.5867 (2008), pp. 1229–1232.
- [60] Zhansong Geng et al. “Graphene Nanoribbons for Electronic Devices”. In: *Annalen der Physik* 529.11 (2017), p. 1700033.
- [61] Li Yang et al. “Quasiparticle energies and band gaps in graphene nanoribbons”. In: *Physical Review Letters* 99.18 (2007), p. 186801.
- [62] Melinda Y Han et al. “Energy band-gap engineering of graphene nanoribbons”. In: *Physical review letters* 98.20 (2007), p. 206805.
- [63] Philip Kim et al. “Graphene nanoribbon devices and quantum heterojunction devices”. In: *2009 IEEE International Electron Devices Meeting (IEDM)*. IEEE. 2009, pp. 1–4.
- [64] Zhihong Chen et al. “Graphene nano-ribbon electronics”. In: *Physica E: Low-dimensional Systems and Nanostructures* 40.2 (2007), pp. 228–232.
- [65] Li Yang et al. “Quasiparticle energies and band gaps in graphene nanoribbons”. In: *Physical Review Letters* 99.18 (2007), p. 186801.
- [66] Martin Evaldsson et al. “Edge-disorder-induced Anderson localization and conduction gap in graphene nanoribbons”. In: *Physical Review B* 78.16 (2008), p. 161407.
- [67] Wentao Xu and Tae-Woo Lee. “Recent progress in fabrication techniques of graphene nanoribbons”. In: *Materials Horizons* 3.3 (2016), pp. 186–207.
- [68] ZF Wang et al. “Tuning the electronic structure of graphene nanoribbons through chemical edge modification: A theoretical study”. In: *Physical Review B* 75.11 (2007), p. 113406.

- [69] Verónica Barone, Oded Hod, and Gustavo E Scuseria. “Electronic structure and stability of semiconducting graphene nanoribbons”. In: *Nano letters* 6.12 (2006), pp. 2748–2754.
- [70] Liang Ma, Jinlan Wang, and Feng Ding. “Recent Progress and Challenges in Graphene Nanoribbon Synthesis”. In: *ChemPhysChem* 14.1 (2013), pp. 47–54.
- [71] Sujit S. Datta et al. “Crystallographic Etching of Few-Layer Graphene”. In: *Nano Letters* 8.7 (2008), pp. 1912–1915. URL: <http://arxiv.org/pdf/0806.3965v1.pdf>.
- [72] Oleg V Yazyev, Rodrigo B Capaz, and Steven G Louie. “Theory of magnetic edge states in chiral graphene nanoribbons”. In: *Physical Review B* 84.11 (2011), p. 115406.
- [73] Hosik Lee et al. “Magnetic ordering at the edges of graphitic fragments: Magnetic tail interactions between the edge-localized states”. In: *Physical Review B* 72.17 (2005), p. 174431.
- [74] Xinran Wang et al. “Room-temperature all-semiconducting sub-10-nm graphene nanoribbon field-effect transistors”. In: *Physical review letters* 100.20 (2008), p. 206803.
- [75] Jian Sun et al. “Lateral plasma etching enhanced on/off ratio in graphene nanoribbon field-effect transistor”. In: *Applied Physics Letters* 106.3 (2015), p. 033509.
- [76] Lingxiu Chen et al. “Oriented graphene nanoribbons embedded in hexagonal boron nitride trenches”. In: *Nature communications* 8 (2017), p. 14703.
- [77] Wan Sik Hwang et al. “Graphene nanoribbon field-effect transistors on wafer-scale epitaxial graphene on SiC substrates”. In: *APL materials* 3.1 (2015), p. 011101.
- [78] Wan Sik Hwang et al. “Transport properties of graphene nanoribbon transistors on chemical-vapor-deposition grown wafer-scale graphene”. In: *Applied Physics Letters* 100.20 (2012), p. 203107.
- [79] Wan Sik Hwang et al. “Room-Temperature Graphene-Nanoribbon Tunneling Field-Effect Transistors”. In: *npj 2D Materials and Applications* 3.1 (2019), pp. 1–7.
- [80] Liying Jiao et al. “Facile synthesis of high-quality graphene nanoribbons”. In: *Nature nanotechnology* 5.5 (2010), p. 321.
- [81] Changxin Chen et al. “Sub-10-nm graphene nanoribbons with atomically smooth edges from squashed carbon nanotubes”. In: *Nature Electronics* (2021), pp. 1–11.
- [82] Kyuhyun Bang et al. “Effect of ribbon width on electrical transport properties of graphene nanoribbons”. In: *Nano convergence* 5.1 (2018), p. 7.
- [83] William Harvey. *Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus*. Latin. 1st. Frankfurt: Sumptibus Guilielmi Fitzeri, 1628.
- [84] Isaiah Berlin. *The Hedgehog and the Fox: An Essay on Tolstoy’s View of History*. English. Ed. by Henry Hardy. 2nd. Princeton University Press, June 2013.